Airbnb Pricing Patterns Across Vancouver Neighbourhoods

Introduction

This project uses Airbnb listing data in Vancouver and analyzes it to identify patterns in the pricing across different neighborhoods, room types, and host status. This is done through a complete end-to-end data analytics pipeline that includes data acquisition, cleaning, storage, analysis, and visualization. The project aims to answer key questions such as how pricing varies by neighborhood, how room type influences pricing, and whether host status affects pricing.

The data used in this project comes from the Inside Airbnb open-source website and includes raw data of individual listings on Airbnb for Vancouver. Important variables in this dataset are the list of the neighborhoods, room type, price, host information, and reviews. There is enough detail in this dataset to do analysis and visualizations.

Data Cleaning and Transformation (ETL Process)

The dataset was cleaned and transformed using Microsoft Excel to prepare it for analysis.

The cleaning process included:

- Selecting relevant variables for analysis.
- Removing duplicate ID entries
- Converting price fields from text into numeric values
- Encoding categorical variables such as host status
- Locating and handling missing values in fields.
- Removed price outliers
- Creating new variables from the data, such as host size grouping

The cleaned dataset was exported as a separate CSV file to highlight the difference between raw and cleaned data in order to maintain data integrity.

Data Storage and Management

A SQLite database was created for storage of the cleaned dataset to allow for the querying and retrieval of the data. The cleaned dataset in the CSV format was loaded into a database table called listings_clean. The data was first checked to ensure corrected data types. Then the SQL queries were executed. This is essential since storing the dataset in the form of a queryable database makes it easier to analyze and scale up the data analysis pipeline.

```
sqlite> PRAGMA table_info(listings_clean);
0,id,TEXT,0,,0
1,host_id,TEXT,0,,0
2,host_since,TEXT,0,,0
3,host_is_superhost,TEXT,0,,0
4,host_listings_count,TEXT,0,,0
5,host_total_listings_count,TEXT,0,,0
6,host_identity_verified,TEXT,0,,0
7,neighbourhood_cleansed,TEXT,0,,0
8,latitude,TEXT,0,,0
9,longitude,TEXT,0,,0
10,room_type,TEXT,0,,0
11,accommodates,TEXT,0,,0
12,bedrooms,TEXT,0,,0
13,beds,TEXT,0,,0
14,price,TEXT,0,,0
15,minimum_nights,TEXT,0,,0
16,availability_365,TEXT,0,,0
17,number_of_reviews,TEXT,0,,0
18,review_scores_rating,TEXT,0,,0
19,reviews_per_month,TEXT,0,,0
20,Superhost_Status,TEXT,0,,0
21,price_per_guest,TEXT,0,,0
22,host_size_group,TEXT,0,,0
```

Data Analysis and Insights

SQL queries were used to analyze the dataset and extract meaningful insights.

The analysis below shows that Airbnb pricing is very different across all Vancouver neighborhoods. The neighborhoods located in the central areas, such as Arbutus Ridge and Downtown, have a higher average price, while areas further away, such as Grandview-Woodland and Fairview, show lower average prices. This analysis shows that location is a primary factor impacting listing prices.

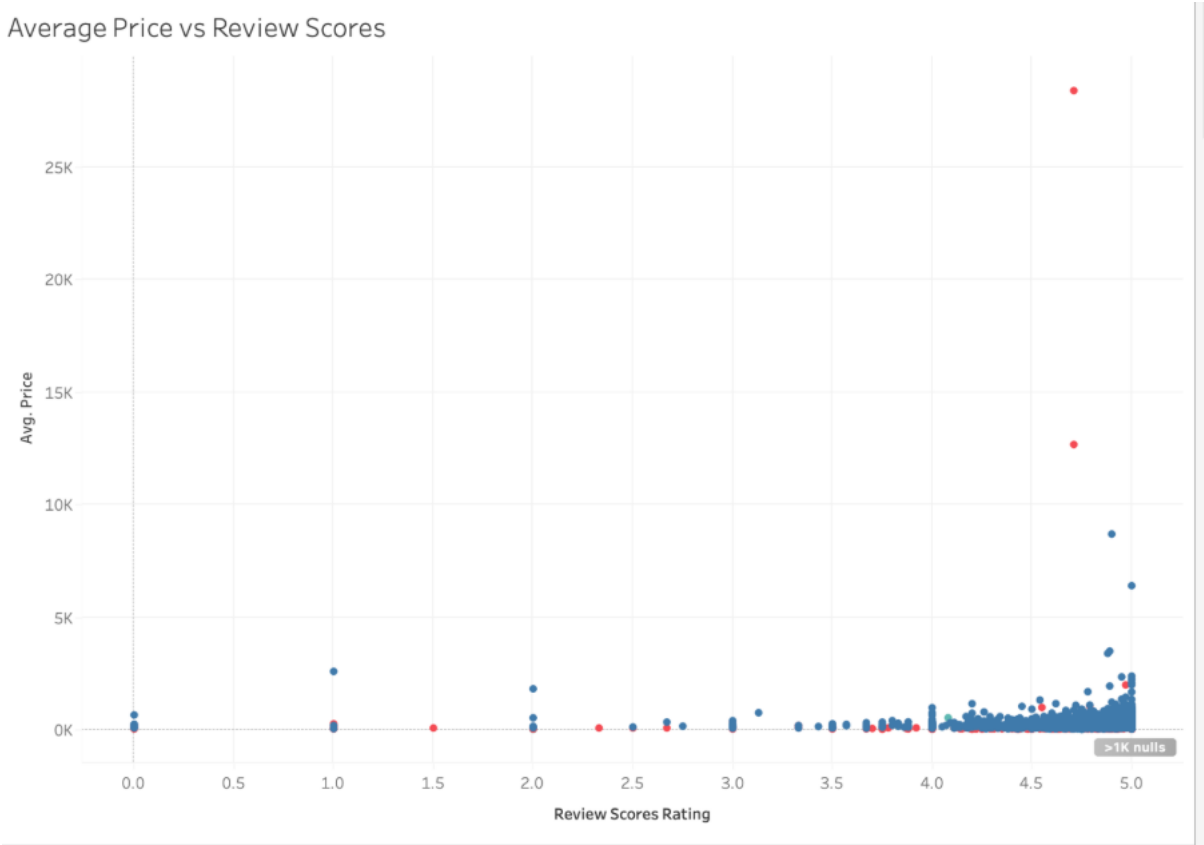
```
sqlite> SELECT
    neighbourhood_cleansed,
    ROUND(AVG(
        CAST(
            REPLACE(REPLACE(price, '$', ''), ',', '')
        AS REAL)
    ), 2) AS avg_price,
    COUNT(*) AS listing_count
FROM listings_clean
GROUP BY neighbourhood_cleansed
ORDER BY avg_price DESC;
"West Point Grey",673.22,565
Marpole,440.01,775
["Arbutus Ridge",359.75,530
Shaughnessy,303.51,365
Downtown,276.11,8555
Kerrisdale,270.12,510
Kitsilano,267.42,2410
"West End",242.5,2510
"Riley Park",231.29,1525
"Mount Pleasant",227.52,2080
Fairview,225.08,765
"Downtown Eastside",224.41,1670
"Dunbar Southlands",217.05,1055
Oakridge,211.33,480
Strathcona,207.87,190
"Kensington-Cedar Cottage",204.73,1890
"South Cambie",199.09,380
Grandview-Woodland,190.99,1330
Sunset,187.25,690
Victoria-Fraserview,184.2,640
Hastings-Sunrise,172.59,1250
Killarney,163.56,445
Renfrew-Collingwood,162.44,1165
neighbourhood_cleansed,0.0,4
sqlite>
```

The room type analysis shows that the entire homes and apartments are priced higher than shared or private rooms. This data proves there is a significant relationship between room type and price.

```
sqlite> SELECT
    room_type,
    ROUND(AVG(
        CAST(
            REPLACE(REPLACE(price, '$', ''), ',', '')
        AS REAL)
    ), 2) AS avg_price,
    COUNT(*) AS listing_count
FROM listings_clean
GROUP BY room_type
ORDER BY avg_price DESC;
"Entire home/apt",269.52,26035
"Private room",165.24,5610
["Hotel room",162.33,15
"Shared room",119.61,115
room_type,0.0,4
sqlite>
```

Host characteristics also influence pricing. Listings associated with super hosts tend to have slightly higher average prices compared to non-super hosts, suggesting that reputation and experience may allow hosts to charge premium rates.

The relationship between price and review scores appears weak. Most listings cluster around high review ratings regardless of price, suggesting that review scores may not strongly impact pricing.



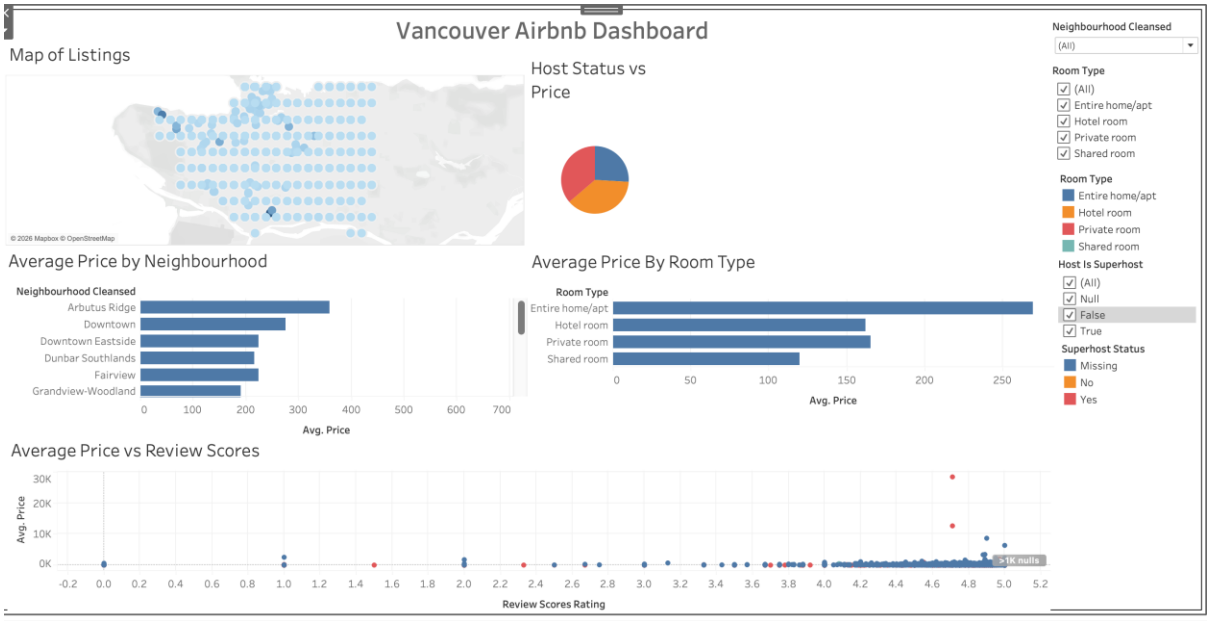
Data Visualization

An interactive Tableau dashboard was created to present the analysis results in a clear and user -friendly format.

The dashboard includes:

- A geographic map showing the different listings across Vancouver's neighbourhoods
- A bar chart showing the average price by neighborhood
- A bar chart showing the average price by room type
- A pie chart of average price based on host status
- Filters for neighbourhood, room type, and host status

These visualizations allow users to explore pricing patterns interactively and gain quick insights into how different factors influence Airbnb pricing.



Limitations

There are a couple of limitations with this analysis. First, it only captures a certain amount of time and doesn't show how the different seasons affect the prices. Secondly, some of the variables had missing values, especially for the reviews. Finally, some prices might be considered outliers due to their high values.

Reproducibility and Project Structure

The project was organized using a structured folder system separating raw data, cleaned data, SQL queries, outputs, and reports. This ensures that the analysis can be reproduced and verified.

The pipeline includes:

- Raw data storage
- Data cleaning in Excel
- Data storage in SQLite
- SQL-based analysis
- Visualization in Tableau

Findings and Conclusions

Based on the results, the factors that affect pricing on Airbnb include location and room type. Location is the most important, as it determines whether prices will be low or high. Listings in the high-demand neighborhoods and downtowns are associated with high pricing, while the same is observed for listings in remote locations. Room type is also another important factor, with whole houses for rent priced higher compared to other types. Superhosts' ability to influence pricing was found to be moderate, while there appeared to be no relationship between review scores and pricing. These results are consistent with the current expectations about Airbnb. This project successfully implemented an end-to-end analytics pipeline to

analyze Airbnb pricing patterns in Vancouver. The results from this project confirm that the key determinants of Airbnb pricing are location and room type.